THE UNIVERSAL SOBER QUOTIENT OF A NOETHERIAN SPACE IS NOETHERIAN SPECTRAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The construction of the universal map from a space to a sober space, which is a bijection on opens commuting with unions and intersections. Invoked inside the counted proof (source lines 4961-4968). Transfers quasi-compactness of opens between the space and its sober quotient.

Source reference. Original lemma lemma-make-sober: sources/stacks-topology/pinned-original-source.tex, source lines 1380–1380. Navigation pointer only: the counted text refers to the Stacks item lemma-make-sober, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a topological space and let $c : X \to X'$ be the universal map from X to a sober topological space, see Lemma lemma-make-sober.*

(1) *If X is quasi-compact, so is X'.*
(2) *If X is quasi-compact, has a basis of quasi-compact opens, and the intersection of two quasi-compact opens is quasi-compact, then X' is spectral.*
(3) *If X is Noetherian, then X' is a Noetherian spectral space.*

Proof. Let $U \subset X$ be open and let $U' \subset X'$ be the corresponding open, i.e., the open such that $c^{-1}(U') = U$. Then U is quasi-compact if and only if U' is quasi-compact, as pulling back by c is a bijection between the opens of X and X' which commutes with unions. This in particular proves (1).

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Proof of (2). It follows from the above that X' has a basis of quasi-compact opens. Since c^{-1} also commutes with intersections of pairs of opens, we see that the intersection of two quasi-compact opens X' is quasi-compact. Finally, X' is quasi-compact by (1) and sober by construction. Hence X' is spectral.

Proof of (3). It is immediate that X' is Noetherian as this is defined in terms of the acc for open subsets which holds for X. We have already seen in (2) that X' is spectral. $\square$